



Species differences in carbon drawdown during marine phytoplankton growth

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ABSTRACT

Ocean alkalinity enhancement (OAE) has been proposed as a mitigation method for negative carbon emission. Its effects on marine phytoplankton communities would depend on species differences in tolerance to high pH, which results from phytoplankton photosynthetic drawdown of dissolved inorganic carbon (DIC). In this study, 20 marine phytoplankton species were grown in sealed batch cultures and DIC, pH and chlorophyll *a* (Chl-*a*) were measured at the peaks of biomass. These results revealed a wide range of species differences. The draw-down DIC (Δ DIC) vs. increases in pH (Δ pH) graph resembled a Michaelis-Menten curve: significantly linear for Δ DIC < ~1000 μ M and starting to plateau at Δ DIC > 1000 μ M. This indicated that two mechanisms were operating: CO₂ limitation at Δ pH < 1.41 and biologically-mediated precipitation-CO₂ released carbon uptake at Δ pH > 1.41. These findings suggest that the potential effects of OAE on the phytoplankton communities would depend on the species differences in oceans.

1. Introduction

Data from the Mauna Loa Observatory in Hawaii, United States of America showed that the atmospheric CO₂ concentration had reached 420 ppm by the end of 2023, an increase of 50 % compared to that in 1850 (280 ppm) before the Industrial Revolution. Oceans take up approximately 26 % of the increased atmospheric CO₂ produced by human activities (Caldeira and Wickett, 2003) and the sustained ocean carbon sink accounted for an overall uptake of ~37 % of the fossil fuel CO₂ emissions from 1850 to 2019 (Friedlingstein et al., 2020), resulting in a pH reduction of 0.1 units in ocean surface waters (Feely et al., 2004; Ma et al., 2023). Annually, 5 Gt CO₂ needs to be removed from the atmosphere by 2050 to keep average global warming well below the 2 °C set by the Paris Agreement (Rogelj et al., 2018; IPCC, 2022), as annual CO₂ emissions have exceeded 40 Gt (Friedlingstein et al., 2022). Among the proposed CO₂ removal strategies, ocean alkalinity enhancement (OAE), which increases the alkalinity of the ocean surface water, has the potential to enhance the natural carbon sink of ocean and alleviate ocean acidification (Poze et al., 2020; Gately et al., 2023; Li et al., 2024).

OAE can be achieved various approaches by adding proton-neutralizing substances to the ocean surface (Renforth and Henderson, 2017). The basic principle of OAE is to cause a shift in the inorganic carbon speciation in seawater, from CO₂ to bicarbonate (HCO₃⁻) and carbonate (CO₃²⁻), creating low-CO₂/high-pH conditions, thereby providing new “space” for absorption of additional atmospheric CO₂ (Hartmann et al., 2013). Ocean acidification (high-CO₂/low-pH) is beneficial to marine phytoplankton (Schaum et al., 2012; White et al., 2020), but OAE conditions (low-CO₂/high-pH) are detrimental to marine phytoplankton (Ferderer et al., 2022). When pulverized minerals or dissolved alkali are added directly to the ocean surface, dissolution products (e.g., trace metals) affect ocean biota immediately (Bach et al., 2019; Foteinis et al., 2023) which is considered an environmental threat for a range of marine ecosystems (Doney et al., 2020). Recent studies have found that the effects of OAE on phytoplankton species are neutral (Gately et al., 2023), positive (Li et al., 2024), or detrimental (Ferderer et al., 2022); however, all these studies demonstrated that OAE significantly effects on phytoplankton community succession. The manner in which marine community compositions change remains unexplored.

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The absence of information on species-specific abundances makes the interpretation of results difficult, further emphasizing the need for species-targeted studies (Ferderer et al., 2022; Gately et al., 2023).

Ocean sequestration of atmospheric CO₂ is an important climate mitigation pathway (Aricò et al., 2021; Dai et al., 2022, 2023), and the amount of CO₂ taken up by marine phytoplankton in continental shelves accounts for as much as 28 % of the total value for oceans (Cai, 2011; Bauer et al., 2013; Dai et al., 2013). According to the Redfield ratio, the absorption of 106 mol of dissolved inorganic carbon (DIC) increases total alkalinity (TA) by 17 mol (Cai et al., 2011); thus, marine phytoplankton photosynthesis can significantly increase the pH, with algal blooms occurring at pH 9.5 in natural aquatic systems (Hansen, 2022) and at pH > 10 in laboratory experiments (Ouyang et al., 2018; Zhang et al., 2019). High pH, triggered by marine phytoplankton photosynthesis, as with OAE, promotes the conversion of DIC to HCO₃⁻ and CO₃²⁻. Therefore, studies on algal growth under low-CO₂/high-pH conditions created by algal photosynthesis would provide a new perspective for examining the effects of OAE on species-specific algae and phytoplankton community succession. Moreover, this approach avoids the interference caused by imported trace elements imported from the dispersal of pulverized minerals.

Ocean acidification and eutrophication jointly influence phytoplankton communities in coastal waters (Gao et al., 2012; Cloern et al., 2014; Riebesell et al., 2018; Dai et al., 2023). Most algae have evolved a physiological CO₂ concentrating mechanism (CCM) to increase intracellular CO₂ concentration and enhance photosynthetic efficiency (Giordano et al., 2005; Reinfeldt, 2011; Beardall and Raven, 2020). The high diversity of algal CCMs results in large species differences in their carbon uptake potential (Raven et al., 2012; Hennon et al., 2017). Following implementation of OAE projects, marine phytoplankton communities may transition from being subjected to ocean acidification to being subjected to alkalization. In this process, the carbon uptake potential of marine phytoplankton and their tolerance to low-CO₂/high-pH may be the key to determining community succession and whether species-specific algae can become the dominant species. The objective of this study was to examine the changes in chlorophyll *a* (Chl-*a*), DIC and pH during the growth of 20 species of marine algae in laboratory batch cultures and compare the species differences in their carbon uptake potential under low-CO₂/high-pH conditions. This will help assess the potential effects of OAE on marine phytoplankton communities.

2. Materials and methods

2.1. Experimental algal species and culture conditions

The 20 algal species used in the culture experiments belonged to 5 phyla (Table 1): 7 chlorophyta, 6 chrysophyta, 4 diatoms, 2 xanthophyta, and 1 dinoflagellate. F/2 medium was added to aged natural seawater (salinity: 31 ± 1) after filtration through a 0.22 μm membrane. Concentrations of main nutrients were 883 μM NaNO₃-N, 36.3 μM NaH₂PO₄-P, and 107 μM NaSiO₃-Si. Before the experiment, 20 algal seeds were activated and grown four times to the logarithmic growth phase. Algae in the logarithmic growth phase were inoculated into 50-mL glass tubes at an initial concentration of 10⁴ cells/mL, and six replicates of the cultures in sealed tubes were set for each algal species. The temperature of the culture room was 21 ± 1 °C, and the light intensity was 78.12 μEinstein m⁻² s⁻¹. In-vivo fluorescence was measured every 12 h. Culture experiments were terminated when the increase in in-vivo fluorescence was stable (<5 % change) for three consecutive readings (stationary phase).

2.2. Sampling and analysis

In the batch cultures experiment, in-vivo fluorescence of 20 algal species was measured every 12 h using a TD 700 fluorometer (Turner Designs, USA). Chl-*a*, pH and DIC were measured at the beginning and

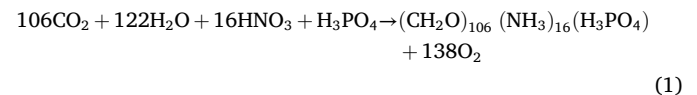
Table 1
Cultured algal species.

Phylum	Algae species	No.
Chlorophyceae	<i>Chlorella</i> sp.	Chl-1
	<i>Dunaliella</i> sp.	Chl-2
	<i>Platymonas subcordiformis</i>	Chl-3
	<i>Pyramimonas</i> sp.	Chl-4
	<i>Tetriselmis</i> sp.	Chl-5
	<i>Platymonas helgolandica</i>	Chl-6
	<i>Dunaliella tertioleta</i>	Chl-7
Chrysophyceae	<i>Emiliana huxleyi</i>	Chr-1
	<i>Phaeocystis globosa</i>	Chr-2
	<i>Prymnesium patelliferum</i>	Chr-3
	<i>Nannochloropsis oculata</i>	Chr-4
	<i>Isochrysis galbana</i>	Chr-5
	<i>Dicrateria zhanjiangensis</i>	Chr-6
	<i>Nitzschia closterium</i>	Dia-1
Diatom	<i>Phaeodactylum tricornutum</i>	Dia-2
	<i>Thalassiosira weissfugii</i>	Dia-3
	<i>Skeletonema costatum</i>	Dia-4
Raphidophyceae	<i>Heterosigma akashiwo</i>	Rap-1
	<i>Fibrocapsa japonica</i>	Rap-2
Dinoflagellate	<i>Amphidinium</i> sp.	Dino

end of the experiment. For Chl-*a*, 50-mL water samples were filtered through 0.7-μm GF/F filters (Whatman, UK) which were then stored in the dark at -20 °C. Chl-*a* concentrations were estimated by fluorometry after acetone extraction overnight at 4 °C (Knap et al., 1996). DIC was measured using a total carbon analyzer (TOC-V_{CPH}, Shimadzu, Japan) after filtration through a 25-mm 0.7-μm GF/F filter, with a mixed carbonate/bicarbonate inorganic carbon standard in an IC reactor. Samples were acidified with hydrochloric acid to a pH of 3 or lower according to the TOC-V_{CPH} manufacturer. pH was measured using a pH meter (DELTA 320, Mettler Toledo, Switzerland) calibrated to seawater pH standard, and precision was ±0.01 pH.

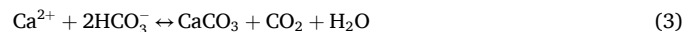
2.3. Data processing and simulation analysis

According to the Redfield ratio, the absorption of 106 mol CO₂ via phytoplankton photosynthesis consumes 16 mol of nitrogen and 1 mol of phosphorus (Eq. 1). The reduction of 17 mol [H⁺] increased TA by 17 mol (Eq. 2).



$$\text{TA} = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] + [\text{B}(\text{OH})_4^-] + [\text{OH}^-] + [\text{HPO}_4^{2-}] + 2[\text{PO}_4^{3-}] + [\text{Si}(\text{OH})_3^-] + [\text{NH}_3] + [\text{HS}^-] - [\text{H}^+]_{\text{F}} - [\text{HSO}_4^-] - [\text{HF}] - [\text{H}_3\text{PO}_4] \quad (2)$$

Calcium carbonate precipitation can be described by Eq. (3).



Theoretically, 1 mol of carbonate precipitation reduces DIC by 1 mol and TA by 2 mol, while releasing 1 mol of CO₂ according to Eq. 3. In practice, the amount of CO₂ released into water per mole of carbonate precipitation is related to pH, temperature, and salinity. For example, only 0.6 mol of CO₂ is released per mole of carbonate precipitation under typical ocean conditions (Soetaert et al., 2007). EXCEL CO2SYS (Pierrot et al., 2011) was used to calculate TA and CO₂ from the measured pH and DIC values based on the constants of Millero (2010).

Origin 2021 was used for data processing and graph drawing, and Pearson's correlation analysis was performed (*p* < 0.05 significance level).

3. Results

3.1. DIC

DIC concentrations initially and at the biomass peaks for 20 marine algae showed large species differences with respect to drawdown (Fig. 1). Based on the ranking of the remaining DIC, *Emiliania huxleyi* (Chr-1) consumed the least amount of DIC (69.23 μM), and *Dunaliella tertioleta* (Chl-7) consumed the largest amount of DIC (1811.67 μM), accounting for 95.2 % of the initial value (Fig. 1). The DIC drawdown was <1000 μM for 8 species and >1300 μM for 10 species. Species differences did not appear to have a specific pattern in relation to functional groups. The largest DIC drawdown of one species was about approximately 26-times higher than that of the species with the smallest DIC drawdown.

3.2. pH

The ranking of maximal pH at the biomass peaks of 20 marine algae differed from that of the DIC drawdown (Fig. 2). Regarding species differences in ΔDIC , maximal pH for *E. huxleyi* (Chr-1) was the lowest, at 8.72 with 0.55 ΔpH (peak-initial), and that for *D. tertioleta* (Chl-7) was the highest, at 10.07 with 1.90 ΔpH . The maximal pH of *Amphidinium* sp. and *Prymnesium patelliferum* was 9.80 and 9.89, respectively, with ΔpH of 1.63 and 1.72, respectively. The maximal pH of the 20 marine algae showed obvious species differences, with the highest ΔpH being approximately 3.5-times higher than the lowest.

3.3. Chl-a

Due to the low initial concentration, the increased Chl-a concentrations ($\Delta\text{Chl-a}$) were approximately equal to the peak concentrations for each species. The ranking of peak Chl-a concentrations of 20 marine algae (Fig. 3) mostly differed from those of DIC drawdown and maximal pH, with *E. huxleyi* having the lowest (52.54 $\mu\text{g/L}$), and *Fibrocapsa japonica* reaching 362.72 $\mu\text{g/L}$. Species differences in peak Chl-a concentrations were 6.9-fold different between the highest and lowest, exceeding 150 $\mu\text{g/L}$ in 17 species.

The ratios of drawdown DIC (ΔDIC , $\mu\text{g/L}$) to chlorophyll *a* (Chl-a, $\mu\text{g/L}$) (C:Chl-a) in the biomass peaks of 20 marine algal species (Fig. 4) ranged from 14.2:1–98.8:1. The C:Chl-a ratios of *Heterosigma akashiwo* (Rap-1) and *E. huxleyi* (Chr-1) were the lowest at 14.2:1 and 15.8:1, respectively, and that of *Platymonas helgolandica* (Chl-6) was the highest at 90.8:1.

3.4. TA

TA of 20 marine algae calculated by CO2SYS using the measured pH and DIC at the biomass peak showed two patterns: 9 species with higher TA, and 11 species with lower TA than the initial TA (Fig. 5). TA of *Thalassiosira weissflogii* and *Dunaliella* sp. reached 2735.66 and 2731.88 μM , respectively, with increases of 529.10 and 525.36 μM to the initial, respectively. TA of *F. japonica* and *D. tertioleta* reached 949.66 and 915.00 μM , respectively, with decreases of 1256.84 and 1291.50 μM to the initial, respectively.

3.5. ΔpH vs. ΔDIC

The correlation between ΔpH vs. ΔDIC of 20 marine algae was fitted to a Michaelis-Menton hyperbolic curve (Fig. 6A). According to the fitted equation, the maximum ΔpH was 2.09 when pH reached 10.26. The half-saturation constant K_m (DIC) of the Michaelis-Menten equation for the 20 species was 392.70 μM . Peak ΔDIC of *E. huxleyi* (69.23 μM) and *H. akashiwo* (243.68 μM) were below the K_m (DIC) of the 20 species.

The hyperbolic curve consisted of a more linear portion for ΔDIC < ~1000 μM before levelling off to reach a plateau for ΔDIC > 1000 μM . The linear regression (10 species) was significant ($p < 0.05$) (Fig. 6B), indicating that consumption of 100 μM of DIC resulted in an increase in pH to 0.112. When ΔDIC was >~800 μM or ΔpH > ~1.41 (pH $\approx$ 9.58), the increase in pH approached the plateau, but was more variable.

3.6. Ambient CO_2 concentrations

Ambient CO_2 concentrations at the biomass peaks of 20 algae studied were calculated using CO2SYS (Fig. 7A) with measured pH, DIC and empirical parameters. Generally, as ΔDIC increased, the remaining CO_2 concentration in the water decreased. Ambient CO_2 concentrations were 0.07 and 0.03 μM , respectively, when peak ΔDIC was 806.24 and 1043.87 μM for *Isochrysis galbana* (Chr-5) and *Amphidinium* sp. (Dino), respectively. Ambient CO_2 concentrations in *E. huxleyi* (Chr-1) and *H. akashiwo* (Rap-1) were >1 μM .

Using the linear relationship between ΔpH and ΔDIC (Fig. 6B) to obtain pH and DIC, CO_2 concentrations were calculated using CO2SYS to determine the correlation between CO_2 and ΔDIC (Fig. 7B). When ΔDIC = 1000 and 1100 μM , the CO_2 concentration was 0.16 and 0.098 μM , respectively, and approached zero with a larger ΔDIC .

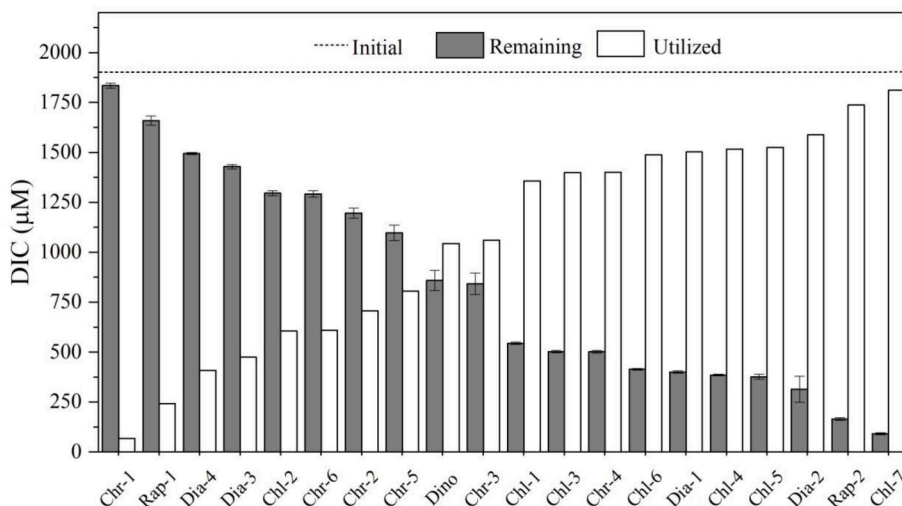


Fig. 1. Remaining and utilized dissolved inorganic carbon (DIC) in biomass peaks during growth of 20 marine algal species. Error bars indicate standard deviation.

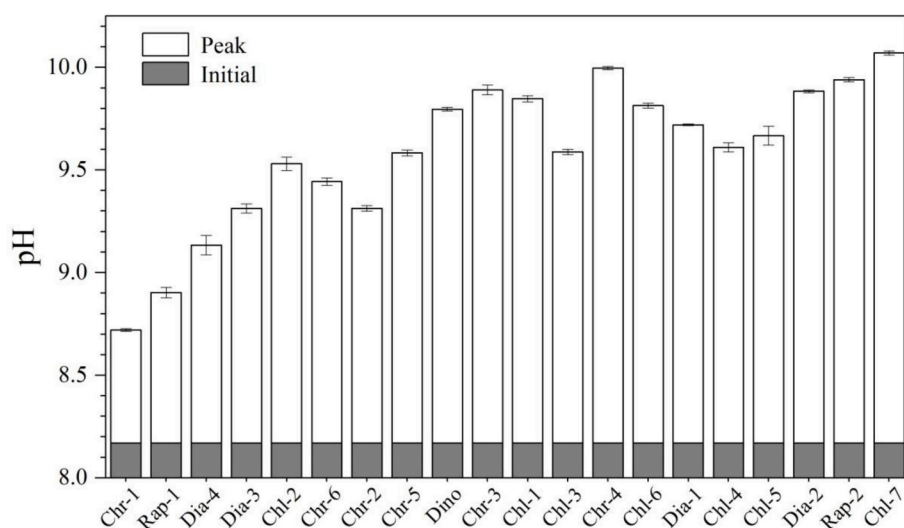


Fig. 2. pH in biomass peaks during growth of 20 marine algal species. Error bars indicate standard deviation.

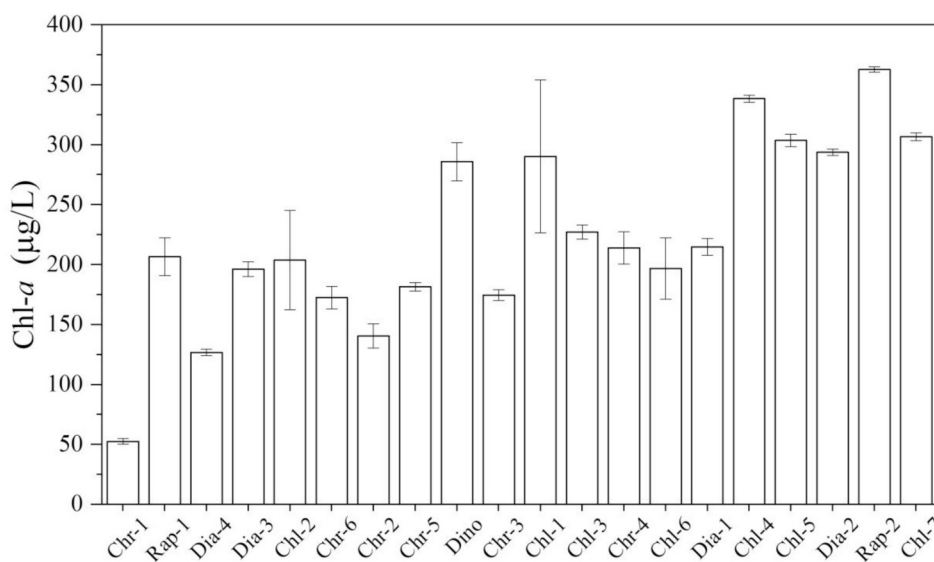


Fig. 3. Chlorophyll a (Chl-a) concentrations in biomass peaks during growth of 20 marine algal species. Error bars indicate standard deviation.

4. Discussion

4.1. Coupling between DIC, Chl-a, and pH during algal growth

Marine algae convert inorganic carbon into organic carbon via photosynthesis, and the ratio of algal cell carbon content to Chl-a (C:Chl-a) is affected by factors such as light, temperature and nutrients (Baumert and Petzoldt, 2008; Wang et al., 2009; Jakobsen and Markager, 2016). Previous studies have found that values of C:Chl-a values for different algae vary from 20 to 120, with an empirical value of 50 (Cloern et al., 1995). In this study, using Δ DIC as the cell carbon content, there was a significant positive correlation between Δ DIC and Chl-a ($p < 0.05$), with the slope of 65.33 ± 12.06 , which is generally consistent with the previous studies.

Photosynthetic utilization of DIC and nutrients by algae increases pH and alters the carbonate system (Hansen, 2022). Previous studies have shown that chemical precipitation intervenes under low- CO_2 /high-pH conditions resulting from high algal photosynthesis (Ouyang et al., 2018; Zhang et al., 2019). When the pH reaches a certain threshold, chemical precipitation occurs and releases CO_2 which can relieve the

carbon limitation of algal photosynthesis (Zhang et al., 2019). In the present study, the hyperbolic curve between Δ pH vs. Δ DIC (Fig. 5B) suggested such regulatory mechanisms for carbon uptake existing in the 20 species, and TA changes also supported this inference. TA concentrations of the 20 marine algae (Fig. 5) showed that the final TA for 8 species with peak Δ DIC $< 1000 \mu\text{M}$ were greater than the initial TA, indicating that CO_2 supply for algal photosynthesis during the peak was completely biologically driven, according to Eqs. (1) and (2). In contrast, the final TA for the 10 species with peak Δ DIC $> 1100 \mu\text{M}$ were all lower than the initial TA, indicating that carbonate precipitation had already occurred (Eq. (3)). For the two algal species with peak Δ DIC from 1000 to $1100 \mu\text{M}$ (i.e., *Amphidinium* sp., and *P. patelliferum*), the theoretical TA should be much larger than the calculated values if no carbonate precipitation occurred, according to Eqs. (1) and (2). Therefore, *Amphidinium* sp. and *P. patelliferum* growth had already induced carbonate precipitation.

In this study, the remaining DIC of the 12 species with peak Δ DIC $> 1000 \mu\text{M}$ (e.g., *D. tertioleta*) reduced almost to zero (Fig. 1), indicating the “high pH-carbonate precipitation-releasing CO_2 ” mechanism could greatly promote the carbon drawdown of marine phytoplankton. A

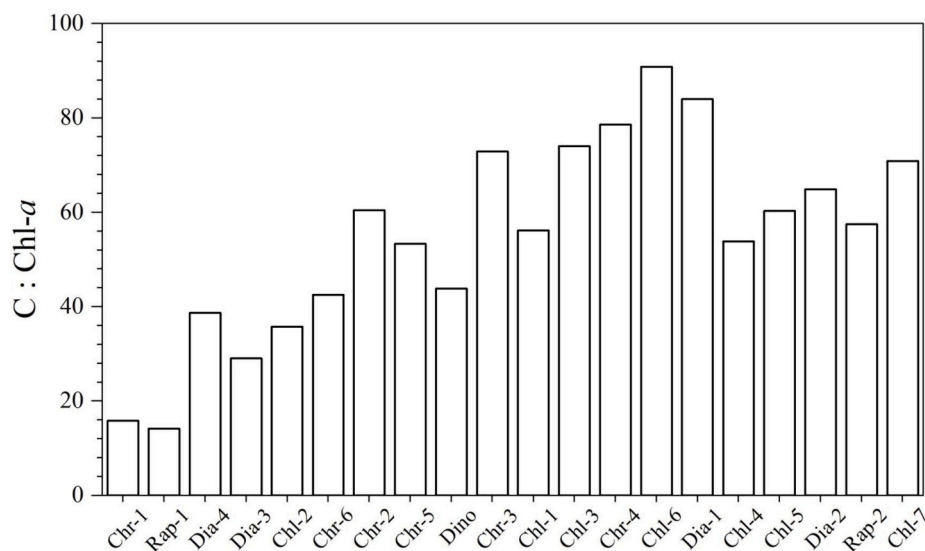


Fig. 4. Ratios of dissolved inorganic carbon (DIC) drawdown (Δ DIC, μ g/L) to chlorophyll *a* (Chl-*a*, μ g/L) (C:Chl-*a*) in biomass peaks during growth of 20 marine algal species.

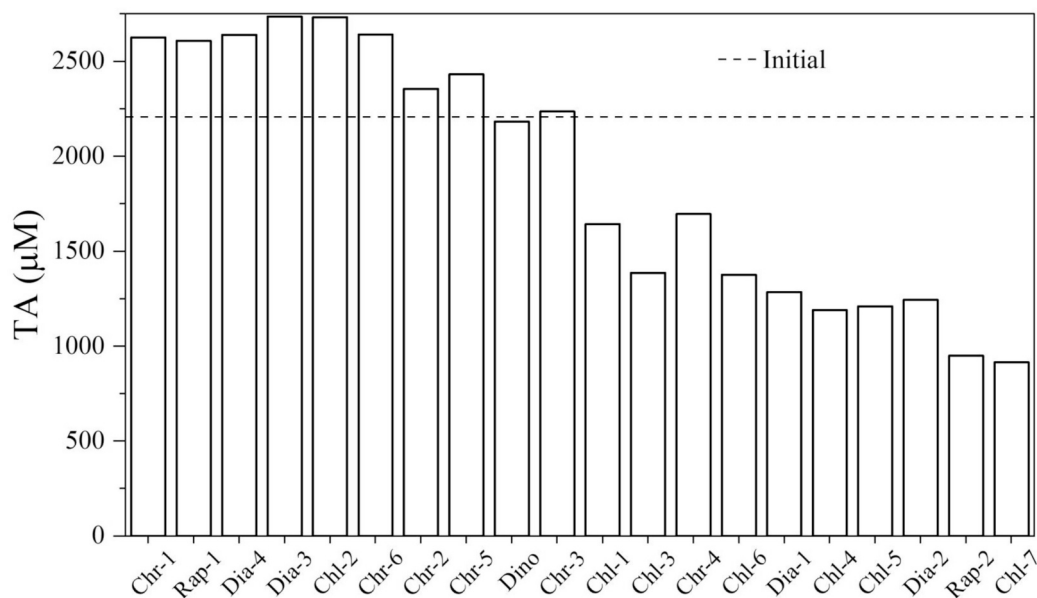


Fig. 5. Total alkalinity (TA) of 20 marine algal species calculated by CO2SYS using measured pH and dissolved inorganic carbon (DIC) in the biomass peaks.

similar scenario has been found in previous studies (Gately et al., 2023); however, the formation of precipitates, resulting from the low CO_2 and high pH and CO_3^{2-} at the end of the *Chaetoceros* sp. experiment, was treated as an undesirable side effect of OAE.

4.2. Species differences in carbon uptake potential

When the cellular CO_2 concentration is low, CCMs increase the CO_2 concentration at the RuBisCO active site by transporting CO_2 and HCO_3^- into cells to maintain algal photosynthesis (Giordano et al., 2005; Reinfelder, 2011; Beardall and Raven, 2020). The apparent CO_2 half-saturation constant of RuBisCO $K_{1/2}(\text{CO}_2)$ is commonly used to compare the CO_2 affinity of CCMs of different algal species (Reinfelder, 2011). Studies have shown species differences in the ability of CCMs, but not differences between algal phyla (Giordano et al., 2005; Reinfelder, 2011). Studies on green algae, diatoms, dinoflagellates, and golden algae show that $K_{1/2}(\text{CO}_2)$ is generally $>0.1 \mu\text{M}$ (Aizawa and Miyachi,

1986; Munoz and Merrett, 1989; Chen and Gao, 2003), but the $K_{1/2}(\text{CO}_2)$ of *Phaeodactylum tricornutum* is as low as $0.075 \mu\text{M}$ (Raven and Johnston, 1991). Generally, CO_2 concentration from 0.075 to $0.1 \mu\text{M}$ in water can be treated as the threshold interval of CCMs. According to the ambient CO_2 concentrations in the biomass peak of the 20 algae species (Fig. 7), ambient CO_2 concentrations would be $<0.1 \mu\text{M}$ when Δ DIC was $>1000 \mu\text{M}$. Thus, whether one algal species can induce $>1000 \mu\text{M}$ of Δ DIC can be used as a simple, rapid criterion for determining whether it has both CCMs and biologically-mediated carbon-fixation mechanisms. Depending on whether algae species have CCMs and biologically-mediated carbon-fixation mechanisms, carbon uptake potential of the 20 marine algal species can be divided into three groups (Table 2).

Group one: two algal species with low carbon uptake potential: *E. huxleyi* and *H. akashiwo*. For these two algal species, DIC uptake was $<K_m(\text{DIC})$ and ambient CO_2 concentrations were $>1 \mu\text{M}$, suggesting that they did not evolve the CCMs.

Group two: six algal species with moderate carbon uptake potential:

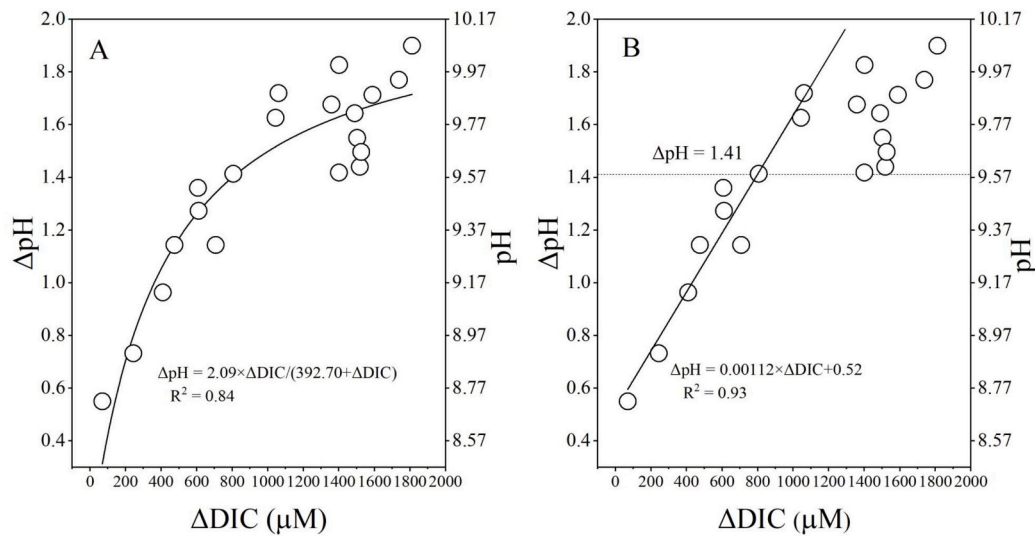


Fig. 6. Correlations between increased pH (ΔpH) and dissolved inorganic carbon drawdown (ΔDIC) in biomass peaks during growth of 20 marine algal species.

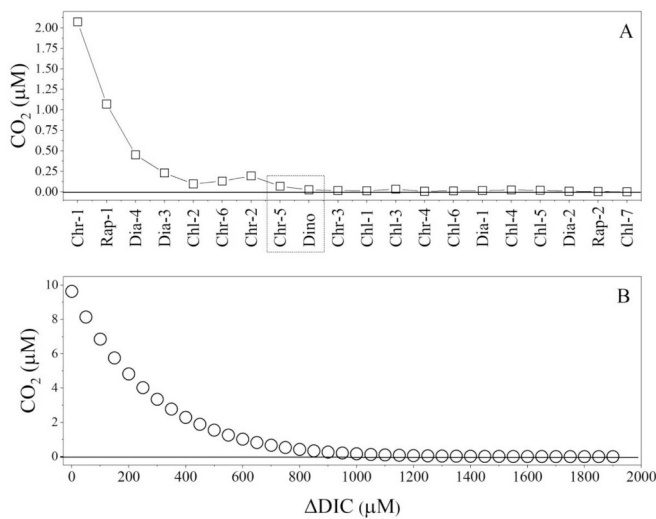


Fig. 7. (A) CO_2 concentrations in biomass peaks during growth of 20 marine algal species; (B) simulated relationship between CO_2 vs. dissolved inorganic carbon drawdown (ΔDIC).

Skeletonema costatum, *T. weissfilgii*, *Dunaliella* sp., *Dicrateria zhanjiangensis*, *Phaeocystis globosa*, and *I. galbana*. For these six species, DIC uptake was $<1000 \mu\text{M}$ and ambient CO_2 concentrations were $>0.07 \mu\text{M}$, with $\Delta\text{pH} < \sim 1.41$, suggesting that they have evolved CCMs; however, the increase in pH due to algal growth was not sufficiently high to trigger carbonate precipitation.

Group three: 12 algal species with high carbon uptake potential: *P. patelliferum*, *Amphidinium* sp., *Chlorella* sp., *Platymonas subcordiformis*, *Nannochloropsis oculata*, *P. helgolandica*, *Nitzschia closterium*, *Pyramimonas* sp., *Tetriselmis* sp., *P. tricornutum*, *F. japonica*, and *D. tertiolela*. For these 12 species, DIC uptake was $>1000 \mu\text{M}$ and ambient CO_2 concentrations were $<0.07 \mu\text{M}$, with $\Delta\text{pH} > \sim 1.41$, suggesting both CCMs and biologically-mediated carbon-fixation mechanisms.

4.3. Implications for responses of phytoplankton species to OAE

Differences in carbon uptake potential of marine phytoplankton species within the same algal phyla (Table 2) increase the complexity of the ecological effects of OAE. Gately et al. (2023) investigated the effects of OAE (limestone-inspired alkalinity additions) on two phytoplankton

Table 2

Carbon uptake potential of 20 marine algal species.

Phylum	Algal species	No.	Carbon uptake potential
Chlorophyceae	<i>Chlorella</i> sp.	Chl-1	+++
	<i>Dunaliella</i> sp.	Chl-2	++
	<i>Platymonas subcordiformis</i>	Chl-3	+++
	<i>Pyramimonas</i> sp.	Chl-4	+++
	<i>Tetriselmis</i> sp.	Chl-5	+++
	<i>Platymonas helgolandica</i>	Chl-6	+++
	<i>Dunaliella tertiolela</i>	Chl-7	+++
	<i>Emiliania huxleyi</i>	Chr-1	+
	<i>Phaeocystis globosa</i>	Chr-2	++
	<i>Prymnesium patelliferum</i>	Chr-3	+++
Chrysophyceae	<i>Nannochloropsis oculata</i>	Chr-4	+++
	<i>Isochrysis galbana</i>	Chr-5	++
	<i>Dicrateria zhanjiangensis</i>	Chr-6	++
	<i>Nitzschia closterium</i>	Dia-1	+++
	<i>Phaeodactylum tricornutum</i>	Dia-2	+++
	<i>Thalassiosira weissfilgii</i>	Dia-3	++
Diatom	<i>Skeletonema costatum</i>	Dia-4	++
	<i>Heterosigma akashiwo</i>	Rap-1	+
	<i>Fibrocapsa japonica</i>	Rap-2	+++
	<i>Amphidinium</i> sp.	Dino	+++

+, ++, and +++ indicate low, moderate and high carbon uptake potentials, respectively.

functional groups (*E. huxleyi* and *Chaetoceros* sp.) and found that the growth rate and elemental ratios of both taxa showed a neutral response to limestone-inspired alkalization. In the present study, we also found that *E. huxleyi* had a low carbon uptake potential (Table 2). Li et al. (2024) demonstrated that OAE (olivine dissolution) promoted the growth of two diatoms (*Thalassiosira pseudonana*, and *P. tricornutum*) and one coccolithophore (*Gephyrocapsa oceanica*), with the highly silicified *T. pseudonana* benefiting the most. Ferderer et al. (2022) enclosed a natural plankton community during a spring bloom to investigate the influence of OAE (enriched with NaOH) on phytoplankton communities and found that simulated OAE had significant but generally moderate effects on various groups and more pronounced effects on the diatom community. In the present study, four diatoms (including *P. tricornutum*) had CCMs and two also had biologically-mediated carbon-fixation mechanisms, indicating that they had greater carbon uptake potential under low- CO_2 /high-pH conditions (Table 2). Although our findings are consistent with recent research, there were species differences in the carbon uptake potential of marine phytoplankton within the same algal

phylum (Table 2), indicating that it is difficult to generalize which phylum is more dominant under OAE conditions. However, determining whether the algae can fix $>1000 \mu\text{M}$ DIC at the biomass peak can be used as a simple and rapid method to assess their response to OAE.

4.4. Limitations

Our study verified the existence of carbonate precipitation through the coupled relationship between pH, DIC and TA; however, the precipitation type and quality were not directly measured.

Notably, the 20 marine phytoplankton species were grown separately in sealed batch cultures under the same cultivation conditions, which might not be optimal for every species since there are species different in the optimum growth conditions. Furthermore, species competition may complicate the effects of low- CO_2 /high-pH on phytoplankton communities in natural systems. An assessment of phytoplankton community responses to OAE is urgently needed to improve understanding of the potential effects of its use on ecosystems.

5. Conclusion

By comparing Chl-*a*, DIC and pH during a cultivation experiment involving 20 marine algal species and using simulation calculations, this study confirmed species differences in the carbon uptake potential of these marine phytoplankton species. Whether phytoplankton species can utilize $>1000 \mu\text{M}$ DIC at the biomass peak can be used as a preliminary criterion for evaluating the carbon uptake potential. Species with high carbon uptake potential mainly utilize the CO_2 released from carbonate precipitation induced by high pH and may be able to tolerate and thrive under OAE. Currently, OAE is mainly achieved by artificially changing the chemical environment of phytoplankton growth, which inevitably introduces negative interference. Our experiments created increased pH conditions owing to phytoplankton growth, which can directly demonstrate the effect of low- CO_2 /high-pH on phytoplankton growth. Thus, the methods used in our study provide a new perspective for assessing the ecological effects of OAE on phytoplankton communities.

CRedit authorship contribution statement

Yafeng Zhang: Writing – review & editing, Writing – original draft, Data curation. **Siru Wang:** Methodology, Data curation. **Bin Gou:** Methodology, Data curation. **Qiabin Wang:** Software, Formal analysis. **Minchi Hou:** Data curation. **Xutao Wang:** Validation. **Kedong Yin:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2024.117021>.

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